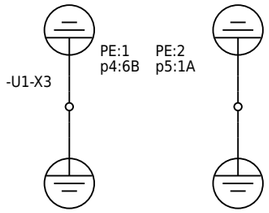
# Cabinet schematic — Pump cabinet

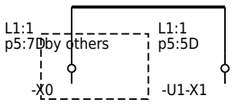
## Pump cabinet

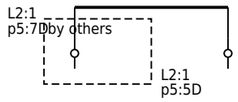
Two direct-on-line motor starters, controlled and monitored by a WAGO 750 PLC through two interface relays.

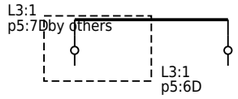
| Revision | Date | Description | Created | Checked | Approved |
|---|---|---|---|---|---|
| 1.1 | 2026-09-01 | First release | SK | | |
| 1.2 | 2026-09-23 | Overload feedback to the PLC | SK | | |
| 1.3 | 2026-10-02 | TeSys D parts, 24V and GND rails, IEC wire colours | SK | | |
| 1.4 | 2026-10-05 | Unit tags, links, busbars and rail bonds, energy on fed supplies | SK | | |
| 1.5 | 2026-10-06 | Three phases on four terminals each, pump n on terminal n+2 | SK | | |
| 1.6 | 2026-10-06 | Owner ruling 2026-10-06, a cabinet is a unit: the place C1 is gone. The change list reads No changes. because the listing drops a place that holds the whole unit, baseline L2, kept by design (REL-DIFF-PLACE) | SK | | |
| 1.7 | 2026-10-16 | Re-cut under fransys 0.19.9, drawings redrawn, no design change | SK | | |
| 1.8 | 2026-10-25 | Pump contact blocks LADN22: the running feedback leaves on the block's second make contact, 83-84 (R780) | SK | | |
| 1.9 | 2026-10-26 | Terminal strip X01 is X4 and breaker F01 is F1: a tag takes no leading zero (R1114) | SK | | |
| 1.10 | 2026-10-28 | No mate, board unit or harness line: the interface relays are wired in the cabinet, and the Ethernet link is a cable (R1216, R1253) | SK | | |
| 1.11 | 2026-11-06 | Re-release on Fransys v0.22.6; the design is unchanged (R1807). | SK | | |

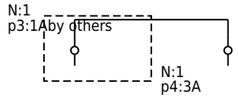
| | | | | | | |
|---|---|---|---|---|---|---|
| Title **Pump cabinet** | | | Number SKX-PC-2 | Revision 1.11 | Revision date 2026-11-06 | |
| Customer | Author SK | Page title Cover | Scope Pump cabinet | Page 1 of 11 | | |

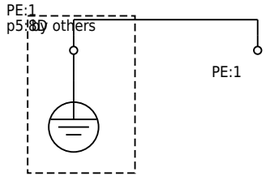


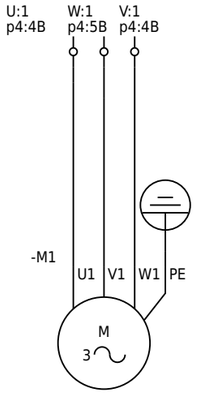



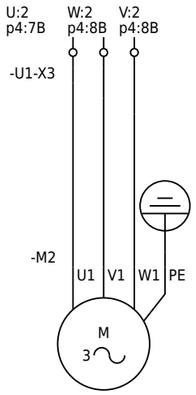



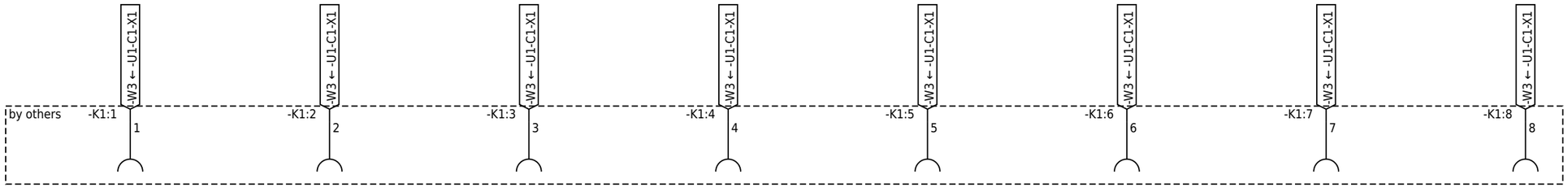



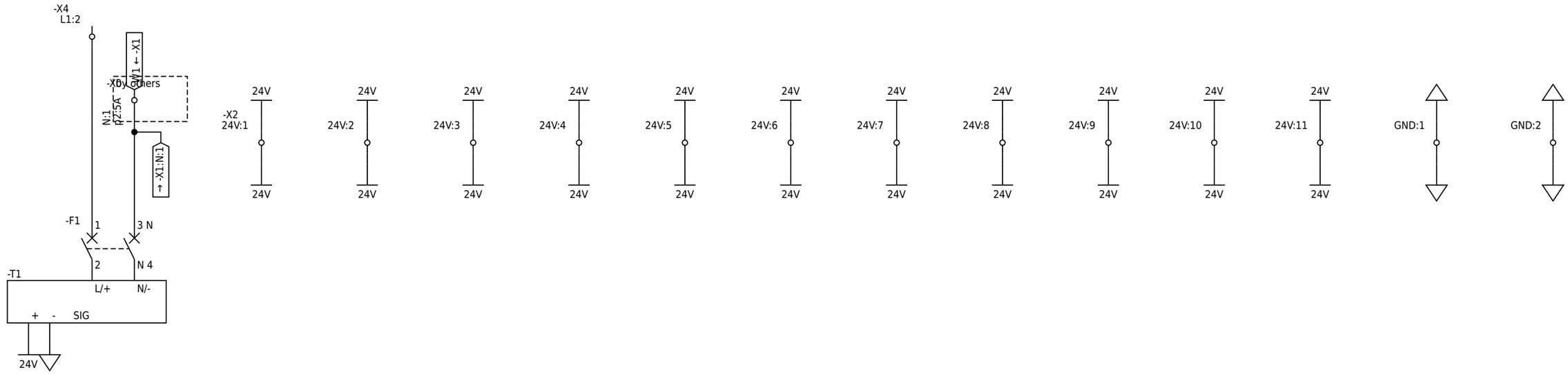



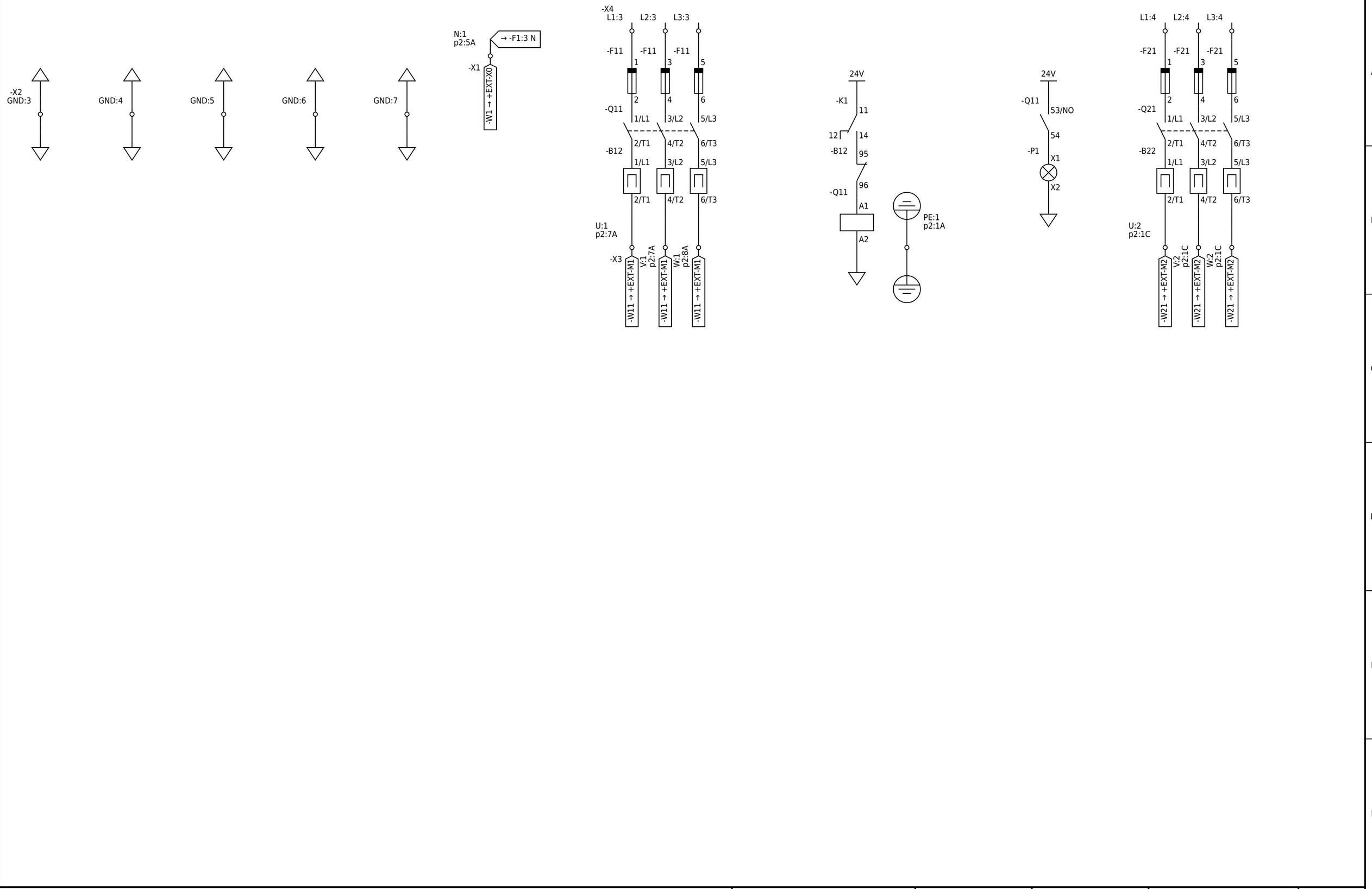



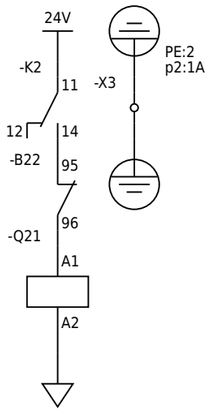

| | | | | | | |
|---|---|---|---|---|---|---|
| Title: **Pump cabinet** | | | Number: SKX-PC-2 | Revision: 1.11 | Revision date: 2026-11-06 | |
| Customer | Author: SK | Page title: Pump 1, Pump 2, Incoming supply, WAGO PLC | Scope: Pump cabinet | Page: 2 of 11 | | |

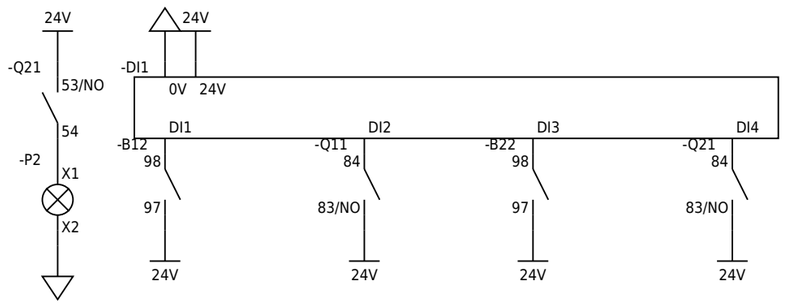


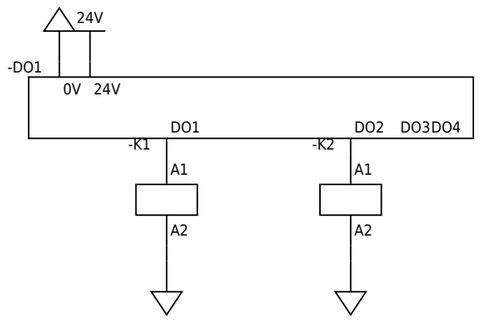

| | | | | | |
|---|---|---|---|---|---|
| Title: **Pump cabinet** | | | Number: SKX-PC-2 | Revision: 1.11 | Revision date: 2026-11-06 |
| Customer | Author: SK | Page title: Control supply, Incoming supply | Scope: Pump cabinet | Page: 3 of 11 | |

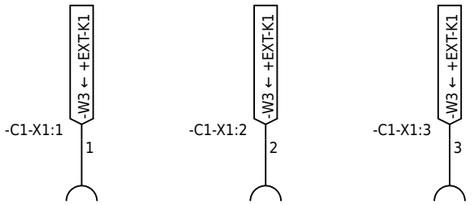



| | | | | | | |
|---|---|---|---|---|---|---|
| Title: **Pump cabinet** | | | Number: SKX-PC-2 | Revision: 1.11 | Revision date: 2026-11-06 | |
| Customer | Author: SK | Page title: Control supply, Pump 1, Pump 2, Incoming supply | Scope: Pump cabinet | Page: 4 of 11 | | |



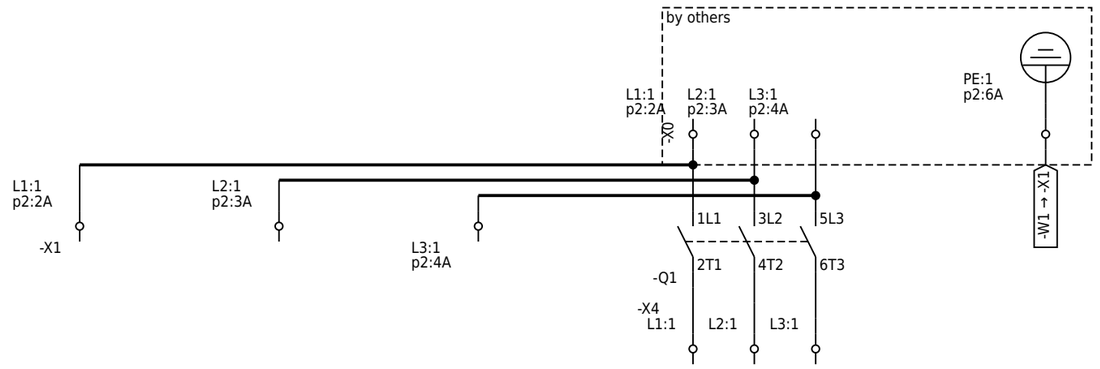



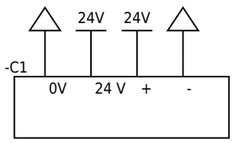










| | | | | | |
|---|---|---|---|---|---|
| Title: **Pump cabinet** | | | Number: SKX-PC-2 | Revision: 1.11 | Revision date: 2026-11-06 |
| Customer | Author: SK | Page title: Pump 1, Pump 2, WAGO PLC, Incoming supply | Scope: Pump cabinet | Page: 5 of 11 | |

# Notes

The upstream board protects the incoming cable with a C16 breaker. The cable lands there on terminals +EXT-X0, the only part of the upstream board in this model.

| | | | | | |
|---|---|---|---|---|---|
| Title: **Pump cabinet** | | | Number: SKX-PC-2 | Revision: 1.11 | Revision date: 2026-11-06 |
| Customer | Author: SK | Page title: Notes | Scope: Pump cabinet | Page: 6 of 11 | |

| Title | Page |
|---|---|
| Pump 1, Pump 2, Incoming supply, WAGO PLC | 2 |
| Control supply, Incoming supply | 3 |
| Control supply, Pump 1, Pump 2, Incoming supply | 4 |
| Pump 1, Pump 2, WAGO PLC, Incoming supply | 5 |

## -DI1

| Pin | Channel name | Signal | Signal name | Wired to |
|---|---|---|---|---|
| -DI1:DI1 | 1 | DI | P1_OVERLOAD | -B12:98 |
| -DI1:DI2 | 2 | DI | P1_RUNNING | -Q11:84 |
| -DI1:DI3 | 3 | DI | P2_OVERLOAD | -B22:98 |
| -DI1:DI4 | 4 | DI | P2_RUNNING | -Q21:84 |

## -DO1

| Pin | Channel name | Signal | Signal name | Wired to |
|---|---|---|---|---|
| -DO1:DO1 | 1 | DO | P1_RUN | -K1:A1 |
| -DO1:DO2 | 2 | DO | P2_RUN | -K2:A1 |
| -DO1:DO3 | 3 | DO | spare | |
| -DO1:DO4 | 4 | DO | spare | |

## -X1

| Designation | Group | Index | Side A | Side B |
|---|---|---|---|---|
| -X1:L1:1 | L1 | 1 | -Q1:1L1 | +EXT-X0:L1:1 |
| -X1:L2:1 | L2 | 1 | -Q1:3L2 | +EXT-X0:L2:1 |
| -X1:L3:1 | L3 | 1 | -Q1:5L3 | +EXT-X0:L3:1 |
| -X1:N:1 | N | 1 | -F1:3 N | +EXT-X0:N:1 |
| -X1:PE:1 | PE | 1 | | +EXT-X0:PE:1 |

## -X2

| Designation | Group | Index | Side A | Bridge |
|---|---|---|---|---|
| -X2:24V:1 | 24V | 1 | -T1:+ | |
| -X2:24V:2 | 24V | 2 | -C1:24 V | |
| -X2:24V:3 | 24V | 3 | -C1:+ | |
| -X2:24V:4 | 24V | 4 | -K1:11 | |
| -X2:24V:5 | 24V | 5 | -K2:11 | |
| -X2:24V:6 | 24V | 6 | -Q11:53/NO | |
| -X2:24V:7 | 24V | 7 | -B12:97 | |
| -X2:24V:8 | 24V | 8 | -Q11:83/NO | |
| -X2:24V:9 | 24V | 9 | -Q21:53/NO | |
| -X2:24V:10 | 24V | 10 | -B22:97 | |
| -X2:24V:11 | 24V | 11 | -Q21:83/NO | |
| -X2:GND:1 | GND | 1 | -T1:- | |
| -X2:GND:2 | GND | 2 | -C1:0V | |
| -X2:GND:3 | GND | 3 | -C1:- | |
| -X2:GND:4 | GND | 4 | -Q11:A2 | |
| -X2:GND:5 | GND | 5 | -P1:X2 | |
| -X2:GND:6 | GND | 6 | -Q21:A2 | |
| -X2:GND:7 | GND | 7 | -P2:X2 | |

## -X3

| Designation | Group | Index | Side A | Side B |
|---|---|---|---|---|
| -X3:PE:1 | PE | 1 | | +EXT-M1:PE |
| -X3:PE:2 | PE | 2 | | +EXT-M2:PE |
| -X3:U:1 | U | 1 | -B12:2/T1 | +EXT-M1:U1 |
| -X3:U:2 | U | 2 | -B22:2/T1 | +EXT-M2:U1 |
| -X3:V:1 | V | 1 | -B12:4/T2 | +EXT-M1:V1 |
| -X3:V:2 | V | 2 | -B22:4/T2 | +EXT-M2:V1 |
| -X3:W:1 | W | 1 | -B12:6/T3 | +EXT-M1:W1 |
| -X3:W:2 | W | 2 | -B22:6/T3 | +EXT-M2:W1 |

## -X4

| Designation | Group | Index | Side A | Bridge |
|---|---|---|---|---|
| -X4:L1:1 | L1 | 1 | -Q1:2T1 | |
| -X4:L1:2 | L1 | 2 | -F1:1 | |
| -X4:L1:3 | L1 | 3 | -F11:1 | |
| -X4:L1:4 | L1 | 4 | -F21:1 | |

## -X4 (cont.)

| Designation | Group | Index | Side A | Bridge |
|---|---|---|---|---|
| -X4:L2:1 | L2 | 1 | -Q1:4T2 | |
| -X4:L2:2 | L2 | 2 | | |
| -X4:L2:3 | L2 | 3 | -F11:3 | |
| -X4:L2:4 | L2 | 4 | -F21:3 | |
| -X4:L3:1 | L3 | 1 | -Q1:6T3 | |
| -X4:L3:2 | L3 | 2 | | |
| -X4:L3:3 | L3 | 3 | -F11:5 | |
| -X4:L3:4 | L3 | 4 | -F21:5 | |

## -C1-X1

PFC200 controller, 2nd generation, 2x ETHERNET, RS-232/-485

| Pin | Ends |
|---|---|
| 1 | +EXT-K1:1 |
| 2 | +EXT-K1:2 |
| 3 | +EXT-K1:3 |
| 4 | +EXT-K1:4 |
| 5 | +EXT-K1:5 |
| 6 | +EXT-K1:6 |
| 7 | +EXT-K1:7 |
| 8 | +EXT-K1:8 |

PFC200 controller, 2nd generation, 2x ETHERNET, RS-232/-485

-C1-X2: no pin wired

| Mpn | Manufacturer | Description | Count | Designations |
|---|---|---|---|---|
| 2002-1201 | WAGO | TOPJOB S feed-through terminal block, 2.5 mm2, grey; 24 A; 800 V | 40 | -X1:L1:1; -X1:L2:1; -X1:L3:1; -X1:N:1; -X2:24V:1; -X2:24V:2; -X2:24V:3; -X2:24V:4; -X2:24V:5; -X2:24V:6; -X2:24V:7; -X2:24V:8; -X2:24V:9; -X2:24V:10; -X2:24V:11; -X2:GND:1; -X2:GND:2; -X2:GND:3; -X2:GND:4; -X2:GND:5; -X2:GND:6; -X2:GND:7; -X3:U:1; -X3:U:2; -X3:V:1; -X3:V:2; -X3:W:1; -X3:W:2; -X4:L1:1; -X4:L1:2; -X4:L1:3; -X4:L1:4; -X4:L2:1; -X4:L2:2; -X4:L2:3; -X4:L2:4; -X4:L3:1; -X4:L3:2; -X4:L3:3; -X4:L3:4 |
| 2002-1207 | WAGO | TOPJOB S ground terminal block, 2.5 mm2, green-yellow | 3 | -X1:PE:1; -X3:PE:1; -X3:PE:2 |
| 3LD2054-0TK51 | Siemens | Switch-disconnector, 3-pole; 16 A | 1 | -Q1 |
| 3NW7033 | Siemens | Fuse holder, 3-pole, 10x38 mm; 32 A | 2 | -F11; -F21 |
| 3NW8004-1 | Siemens | SENTRON cylindrical fuse link, 10x38 mm, aM; 4 A | 6 | -F11; -F21 |
| 3SU1102-6AA40-1AA0 | Siemens | Pilot light 22 mm, green, integrated LED 24 V AC/DC | 2 | -P1; -P2 |
| 40.61.9.024.1000 | Finder | PCB relay, 1 changeover contact, 16 A, 24 V DC coil | 2 | -K1; -K2 |
| 5SY6506-7 | Siemens | Miniature circuit breaker, 1P+N, curve C; 6 A | 1 | -F1 |
| 750-402 | WAGO | 4-channel digital input module, 24 V DC | 1 | -DI1 |
| 750-504 | WAGO | 4-channel digital output module, 24 V DC; 0.5 A | 1 | -DO1 |
| 750-600 | WAGO | End module, terminates the internal data bus; no data contacts | 1 | -E1 |
| 750-8212 | WAGO | PFC200 controller, 2nd generation, 2x ETHERNET, RS-232/-485 | 1 | -C1 |
| LADN22 | Schneider Electric | TeSys D front-mount auxiliary contact block, 2 NO + 2 NC | 2 | -Q11; -Q21 |
| LC1D09BD | Schneider Electric | TeSys D contactor, 3-pole, AC-3 9 A, 1 NO + 1 NC auxiliary, 24 V DC coil | 2 | -Q11; -Q21 |
| LRD08 | Schneider Electric | TeSys thermal overload relay, class 10A, 2.5-4 A, 1 NO + 1 NC | 2 | -B12; -B22 |
| QUINT4-PS/1AC/24DC/2.5/SC | Phoenix Contact | Power supply, 1-phase, 24 V DC; 2.5 A | 1 | -T1 |